

CCPROG3 AY 2025 – 2026 3rd Term MP Specifications
“Yohane The Parhelion! The Siren in the Mirror World! FULL SONG EDITION” [1]



In this game, players will take on the role of Yoshiko Tsushima (otherwise known as “Yohane”), one of the nine members of the school idol group **Aqours**, to solve mysterious phenomena involving idols losing their voices everywhere.

Game Premise

A school idol group known as Aqours consisting of Chika, Riko, You, Hanamaru, Ruby, Dia, Mari, Kanan, and Yoshiko are preparing for a live that will be held in Tokyo along with J-pop idols and K-pop Idols. However, a phenomenon started to occur when these idols started losing their voices.

One day, members of Aqours started losing their voices as well, and Yoshiko (who calls herself Yohane) vows to unleash her inner fallen angel named “Yohane” to solve this mystery. One evening, during a peaceful moment of slumber, she awakens into a mirror world where it is discovered that a siren has been stealing the voices of idols from her dimension in an attempt to have the perfect singing voice. Guided by her companion, Lailaps, she must venture into the

domain of this Siren and rescue her fellow idols' alternate selves in this dimension, retrieve their voices back, and foil the Siren's scheme once and for all!

Game Setting

The game takes place in Numazu, which is located in the Shizuoka prefecture of Japan. In the course of the game, Yoshiko will have to travel to different locations of Numazu in the mirror world that have been taken over by the Siren that is holding the alternate selves of her fellow Aqours' members hostage. These locations are different for each playthrough and depend on the Aqours member that is currently held hostage by the Siren.

Gameplay Mechanics

The game is a rogue-lite game, where each playthrough consists of (possibly) changing environments. The most important aspects for each playthrough is that the gameplay proper follows the following blueprint:

- 1.) At the start of the game, 3 (out of 8) idols are "randomly" selected to be the one that Yohane will rescue. This, in turn, will determine the location name of the 3 dungeons available for that playthrough
- 2.) Each dungeon consists of a 2D environment that contains obstacles and traps that require players to lead Yoshiko and Lailaps through. Reaching the end indicates that a dungeon is "cleared".
- 3.) After a dungeon is cleared, that dungeon is no longer selectable for the duration of that playthrough. Players then have the option to select another dungeon to clear. This process is repeated until the 3 dungeons are cleared, and therefore, the 3 idols held hostage for that playthrough are rescued
- 4.) After the 3 dungeons are cleared, the "final dungeon" becomes available, which is where the Siren is located in
- 5.) Clearing the final dungeon indicates the defeat of the Siren, meaning that the player has won the game!

Main Menu Module (Console Version)

Once you start the application, the player is provided with a main menu that will enable them to either start a new game (if no current playthrough is happening), continue the game (if a playthrough exists), and to view their current achievements. A sample main menu module exists in the figure below which may be used as a reference.

```
*****
*           Yohane The Parhelion!           *
*       The Siren in the Mirror World!       *
*****

[N]ew Game
[S]tatus
[Q]uit

Your choice:
```

If a playthrough is currently happening, [C]ontinue should be an option available as shown below

```
*****
*           Yohane The Parhelion!           *
*       The Siren in the Mirror World!       *
*****

[C]ontinue
[N]ew Game
[S]tatus
[Q]uit

Your choice:
```

Whenever the user has completed a game OR suffered a game over from a previous playthrough, [N]ew Game is replaced with [N]ew Game+

```
*****
*           Yohane The Parhelion!           *
*       The Siren in the Mirror World!       *
*****

[N]ew Game+
[S]tatus
[Q]uit

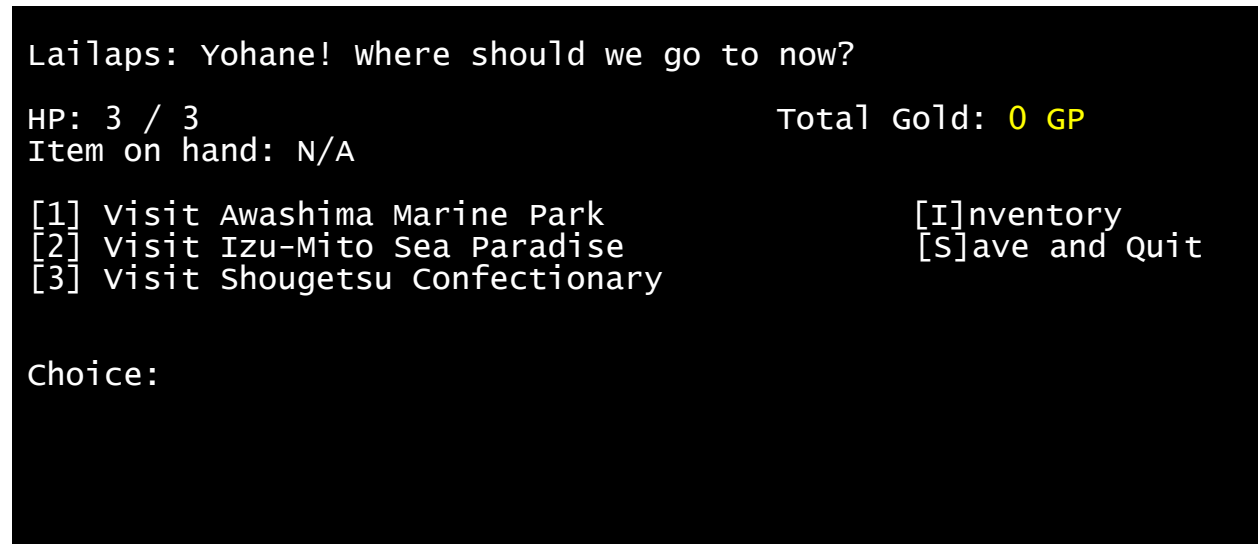
Your choice:
```

More specifically, the following are the functionalities of the options selected:

- 1.) New Game – Begins a new playthrough. Overwrites the current playthrough if a game is currently being played. Does NOT overwrite the status from previous playthroughs earned (e.g. Idols saved, unlocks earned, etc.)
- 2.) Continue – Loads a playthrough that has been saved. Only one playthrough may be saved and loaded at a time
- 3.) New Game+ - Begins a new playthrough, however gold earned from a previously completed playthrough carries over. Items purchased, however, is reset back to 0.
- 4.) Status – View Total Stats of the Player. More on this later
- 5.) Quit – Terminates the Program

Gameplay Interface

At the start of each playthrough, players are informed of which Aqours members have currently been held hostage by the Siren in the Mirror World that Yohane will have to rescue. Afterwards, they are presented with a menu that will allow them to traverse through the 3 dungeons in the order that they wish to proceed in. An example is as follows:



Specifically, the following choices are available in the game menu:

- 1.) Choices 1 – 3 = Traverse a dungeon. Note that a dungeon that has been cleared should have the choice replaced with an x, indicating that it is no longer selectable.
- 2.) Inventory = Selected using the 'I' character. This module allows Yohane to view the items available to her.
- 3.) Save and Quit = Returns to the main menu AND saves your progress.
- 4.) Visit Hanamaru's Shop = Selected using the 'H' character. Available ONLY Once Hanamaru has been rescued in the current playthrough, or in previous playthroughs.

The Dungeon Names are dependent on which character the user will have to save. They are as follows:

Character to Save	Dungeon Name
Chika	Yasudaya Ryokan
Riko	Numazu Deep Sea Aquarium
You	Izu-Mito Sea Paradise
Hanamaru	Shougetsu Confectionary
Ruby	Nagahama Castle Ruins
Dia	Numazugoyotei
Kanan	Uchiura Bay Pier
Mari	Awashima Marine Park

Clearing a dungeon makes that dungeon unselectable for the playthrough. An example menu showing the two dungeons (Awashima and Izu-Mito Sea Paradise) out of three dungeons cleared is seen below

Lailaps: Yohane! Where should we go to now?

HP: 4 / 5

Total Gold: 362 GP

Item on hand: N/A

[X] Visit Awashima Marine Park
[X] Visit Izu-Mito Sea Paradise
[3] Visit Shougetsu Confectionary

[I]nventory
[S]ave and Quit

Choice:

Dungeon Interface (Console Edition)

Once the player has decided to visit a dungeon, the dungeon interface should enable the players a good view of what the dungeon's contents and the player's current state is. A reference is indicated below

```
Dungeon #1: Izu-Mito Sea Paradise
Floor 1 of 2

HP: 3 / 3                               Total Gold: 24 GP
Item on hand: N/A

*****
* . . . . X . . . . . X . . . . . *
* . T . X . . . . h . . . . . X . . . . . b . . . . . *
* . . . . X . . . . hh . . . . . X . . . . . ww . . . . . XX . . . . . *
* . . . . XX . . . . b . . . . . XX . . . . . ww . . . . . XX . . . . . X . . . . . b . . . . . *
* . . . . X . . . . V . . . . . XX . . . . . X . . . . . ww . . . . . XX . . . . . b . . . . . X . . . . . *
* . X . . X . . . . X . . . . . VX . . . . . X . . . . . ww . . . . . XX . . . . . X . . . . . *
* . X . . YX . . . . X . . . . . VX . . . . . ww . . . . . XX . . . . . VVV . . . . . XXXbXX *
* . . X . . VVVVVVVVVVVVVV . . . . VVVVV ww . . . . . XX . . . . . h . . . . . VV . . . . . *
* . . X . . . . X . . . . . . . . . . VVV . . . . . . . . . . h . . . . . VV . . . . . *
* . . . . . X . . . . . . . . . . V . . . . . b . . . . . h . . . . . VV . . . . . b . . . . . E *
*****
```

Specifically, whenever the player decides to traverse a dungeon, the following information should be indicated:

- 1.) Dungeon no. and name of the stage – This is depicted on the dungeon that the player selected
- 2.) Floor no. – Indicates their level within the dungeon
- 3.) HP – The amount of hitpoints. If this reaches 0, it's game over!
- 4.) Item on hand - The item that Yohane currently has with her in the dungeon. (More on this later)
- 5.) Total Gold – Amount of Gold Yohane has on hand
- 6.) Dungeon Layout – A 2D grid representing the contents of the dungeon Yohane is currently traversing

Dungeon Layout (Console Edition)

It is important that the interface for each dungeon contains the following:

- 1.) Yohane's location – Denoted with the character 'Y'. This is where Yohane is in the dungeon. Players have the option to move her with the WASD movement scheme. This means 'W' moves her up, 'A' moves her left, 'S' moves her down, and 'D' moves her right. There is also an option to press 'X' which makes her stay still and not move
- 2.) Dungeon border – Denoted with the character '*'. These are impassable and should surround the dungeon at all times
- 3.) Passable Tiles – Depicted with the character '.'. These are areas that Yohane (and enemies) can pass through without any consequences
- 4.) Wall Tiles – Depicted with the character 'v'. These are wall tiles that Yohane can dig and destroy, replacing them with passable tiles '.'. Do note that, when digging Wall tiles, Yohane stays put and does not advance to the direction of the Wall tile she just dug
- 5.) Spike Tiles – Depicted with the character 'x'. These are tiles that, while Yohane can dig and destroy, will damage Yohane. Destroying a spike tile deals .5 HP damage
- 6.) Water Tile – Depicted with the character 'w'. These are tiles that Yohane cannot pass over, but certain enemies (like Bats) can pass over.
- 7.) Heat Tile – Depicted with the character 'h'. These are tiles that are passable to Yohane, but should she not move while standing on top of one (either through digging, attacking an enemy, or intentionally not moving), she will take 1 HP damage.
- 8.) Treasure Tile – Depicted with the character 'T'. Gives Yohane a random treasure, either gold (Random value between 10g to 100g) or an item (Noppo Bread). Only one Treasure tile spawns per dungeon floor.
- 9.) Exit Tile – Depicted with the character 'E'. This allows Yohane to proceed to the next floor of the dungeon or end the dungeon when they are at the last floor
- 10.) Bats Tile – Depicted with varying characters ('b') in the example. More on this in a later section

The no. of floors of a dungeon denotes the no. of levels or layouts that the player will have to traverse to. They are as follows:

- 1.) 1st Dungeon = 1 floor
- 2.) 2nd Dungeon = 2 to 3 floors
- 3.) 3rd Dungeon = 3 to 4 floors

Moving

When Yohane is traversing a dungeon, everything she does is considered a movement or a turn. This includes things like:

- 1.) Moving Up, Down, Left, or Right
- 2.) Pressing an invalid input (In which case, Yohane will NOT move)
- 3.) Digging Walls
- 4.) Activating her item on hand
- 5.) Switching between items in her possession

In short, anything that involves pressing a button is considered as a movement made by Yohane. Movements are important as different elements within the dungeon will react according to the movement Yohane made during her turn.

Digging Walls

When Yohane comes in contact with a wall tile that she can dig, the wall tile disappears and is replaced with a passable tile. However, Yohane remains still. The demonstration can be seen in the figure below. Here, Yohane is adjacent to a wall tile to her right. This means if the player tells her to move right (presses 'D') while she is to the left of the wall, it counts as "digging" the wall.



Do note that digging a wall counts as a movement / turn. Be careful of digging walls while standing on top of heat tiles as doing so counts as being idle in front of one, leading to damage!

Switching & Using Items

Yohane is able to switch between items that she has on-hand. The following items are the ones Yohane is able to “equip” by cycling through her items in her inventory:

- 1.) Tears of a Fallen Angel
- 2.) Noppo Bread
- 3.) Choco-mint ice cream *

Note that switching between items is counted as a movement, meaning enemies will move despite Yohane being stationary. Likewise, Yohane is able to use an item on-hand except for the Choco-mint ice cream which activates automatically as long as she has it on-hand.

When Yohane does not have any items, it will be indicated as N/A and using an item when there are no items on hand counts as a movement (which acts as an invalid button press).

The default button controls for navigating through the item on-hand are the '[' button for previous item, ']' for the next item, and the space bar for using the item. Likewise, the quantity is displayed next to the item to indicate how many items they have in their inventory.

No item on hand

HP: 3 / 3 Total Gold: 24 GP

Item on hand: N/A

Has 1x Tears of a Fallen Angel on hand

HP: 3 / 3 Total Gold: 24 GP
Item on hand: Tears of a fallen angel

Has 4x Tears of a Fallen Angel

```
HP: 3 / 3                                     Total Gold: 24 GP
Item on hand: Tears of a fallen angel (4)
```

Bats and Attacking

Bats are considered the main minions of the Siren that has taken over the mirror world of Numazu holding Yohane's fellow Aqours members. Therefore, bats are the main enemies that Yohane will have to deal with.

The behavior of bats vary, but they start of simple to deal with and get more complicated as Yohane clears more dungeons during the playthrough. Specifically, the table describes their behavior

1 st Dungeon (Drops 5gp upon defeat)	Bats move AFTER every 2 movements made by Yohane. They can only move Up, Down, Left, or Right. The direction they move is random. However, if they are about to move and Yohane is on an adjacent tile, they will attack Yohane. Bats from the 1st dungeon deal .5 HP of damage.
2 nd Dungeon (Drops 10gp upon defeat)	Bats move during every movement. They can move Up, Down, Left, or Right. The direction they move is random. However, if they are about to move and Yohane is on an adjacent tile, they will attack Yohane. Bats from the 2nd dungeon deal 1 HP of damage.
3 rd Dungeon (Drops 15gp upon defeat)	Bats move during every movement. They can move in any direction, and diagonally. The direction they move is random. However, if they are about to move and Yohane is on an adjacent tile, they will attack Yohane. Bats from the 3rd dungeon deal 1.5 HP of damage.

To give you an illustration of the bats' movement, see the following example of a bat encounter in the 1st dungeon.

1.) In the illustration below, we can see Yohane and a bat nearby.	2.) Yohane moves to the right. As it's the first dungeon, bats only move after every 2 movements. This means that the bat is going to move after Yohane's next movement	3.) Yohane once again moves to the right. Yohane has moved twice now, so the bat now moves. It moves randomly and moves down as a result
<pre> ***** * .xv .b* *Y....* * .X....* ***** </pre>	<pre> ***** * .xv .b* * .Y....* * .X....* ***** </pre>	<pre> ***** * .xv . . * * . .Y .b* * .X....* ***** </pre>
4.) At this point, Yohane will once again move to the right and move adjacent to the bat. Do note that the bat will not move just about yet as it moves after every two moves made by Yohane	5.) Should Yohane press right again, she will "attack" the bat, killing it and making the bat drop gold in it's place. Yohane can then move over the tile and pick up the gold	6.) The bat is gone and Yohane now has more gold in her possession that she can use in Hanamaru's store later on
<pre> ***** * .xv . . * * . . .Yb* * .X....* ***** </pre>	<pre> ***** * .xv . . * * . . .Yg* * .X....* ***** </pre>	<pre> ***** * .xv . . * *Y* * .X....* ***** </pre>

However, be wary of Yohane's movement when multiple bats are nearby. Accidentally jumping into a tile adjacent to a bat that is about to move results in damage. The following gives an explanation of such scenario

1.) In the illustration below, we can see Yohane and there are two bats	2.) Yohane moves to the left. As t's the first dungeon, once again, bats only move after every two movements. That means the bats are going to move after Yohane's next movement	3.) Yohane decides to move left. The bat further away from her moves left. HOWEVER, she ended up being adjacent to a bat below her. The bat then attacks Yohane and deals .5 HP of damage (the attacking bat turns into an uppercase B for that moment signifying it landed an attack). Note that bats that attack stay in place and do not move
<pre> ***** *.xv.b* *..Y..* *bX...* ***** </pre> HP 3 / 3	<pre> ***** *.xv.b* *.Y...* *bX...* ***** </pre> HP 3 / 3	<pre> ***** *.xvb.* *Y....* *BX...* ***** </pre> HP 2.5 / 3
4.) At this point, Yohane can either attack the bat that just damaged her (by pressing down), but the player gets flustered and moves up. The bat that previously attacked reverts to a lowercase b. Once again, the bats don't move as they move after every two movements made by Yohane	5.) The player makes a mistake and ill-advisedly moves down again. As this is a turn when bats can move, the bat adjacent to Yohane lands another attack, damaging her again. Meanwhile, the bat further away from Yohane moves down randomly	6.) The player finally gets their act together and attacks the bat that has been tormenting them by pressing down. The bat is killed and replaced with gold
<pre> ***** *Yxvb.* *.....* *bX...* ***** </pre> HP 2.5 / 3	<pre> ***** *.xv...* *Y..b.* *BX...* ***** </pre> HP 2 / 3	<pre> ***** *.xv...* *Y..b.* *gX...* ***** </pre> HP 2 / 3

7.) Yohane decides against picking up the gold and decides to save it for later. The player then makes her move right. Again, bats will make a move after the next movement	8.) Yohane moves right again, into an incoming bat. The bat attacks Yohane and deals her .5 damage	9.) Yohane gets surprised and ends up pressing 'up' ('W') instead. This leads her to dig the wall above her. However, as bats move after every two movements, the bat does not attack her
<pre> ***** *.XV.* *.Y.b.* HP 2 / 3 *gX.* ***** </pre>	<pre> ***** *.XV.* *..YB.* HP 1.5 / 3 *gX.* ***** </pre>	<pre> ***** *.X.* *..Yb.* HP 1.5 / 3 *gX.* ***** </pre>
10.) Yohane decides to move up, forgetting about the bat. The bat, decides to move randomly again, considering Yohane is not adjacent to it when it makes it's move, so it moves right	11.) Thinking that it's time to make the bat pay, Yohane decides to move right towards the bat. However, as the bat moved in the previous turn, it won't move in this one. However...it will move in the next one	12.) Once again, Yohane ends up moving right, ill-advisedly towards the bat's incoming movement. The bat once again deals .5 damage to Yohane.
<pre> ***** *.XY.* *...b.* HP 1.5 / 3 *gX.* ***** </pre>	<pre> ***** *.X.Y.* *...b.* HP 1.5 / 3 *gX.* ***** </pre>	<pre> ***** *.X..Y* *...B.* HP 1 / 3 *gX.* ***** </pre>
13.) Finally, Yohane decides to put the bat out of it's misery by correctly pressing down, and killing the bat. Gold replaces that bat's location	14.) At this point, Yohane is free to continue exploring. However, watch out for other bats!	Note: If Yohane ends up jumping into a tile being attacked by multiple bats, Yohane is only attacked by one. That means if two bats are able to attack her, only one of them will attempt to attack her.
<pre> ***** *.X..Y* *...g.* HP 1 / 3 *gX.* ***** </pre>	<pre> ***** *.X..Y* *...g.* HP 1 / 3 *gX.* ***** </pre>	

Completing a Dungeon

A dungeon is marked as complete once a player is able to reach the exit. The player is then informed of what is unlocked (in the current playthrough and future playthrough) based on what idol has been rescued. See below for an example of a Victory Screen:

Dungeon Cleared!
Shougetsu Confectionary Completed!
Hanamaru Kunikida rescued!

Unlocked: Hanamaru's Store Now Available!

Hanamaru: Yohane-chan, zura! You're here!

Yohane: Hanamaru! We have to get out of here quickly!

Hanamaru: Oh? I was wondering what this place was and why there are bats everywhere, zura!

Yohane: Seems like there's a siren that wants to take your voices and is holding you in this dimension so that your counterparts in the real world can't sing!

Hanamaru: Really? That sounds terrifying, zura. What have we got to do?

Yohane: First, we have to get out of here, Zoramaru! I know the way out.

Hanamaru: Lead the way, zura!

[R]eturn

Once an idol has been rescued, the item that unlocks upon rescuing them will be available in subsequent playthroughs.

Hanamaru's Store

In the mirror world, Hanamaru acts as a merchant who specializes in selling sweets, but is able to sell varying items to Yohane the more Aqours members she rescues.

The following are items that Yohane is able to purchase from Hanamaru

Item Name	Price	Description	Availability
Tears of a fallen angel	30GP	Heals Yohane .5 HP	Available at the start. Can only be purchased once for that playthrough.
Noppo Bread	100GP	Heals Yohane .5 HP	Available at the start.
Shovel Upgrade	300GP	Allows Yohane to dig spike walls without damage	Rescue Kanan. Can only be purchased once for each playthrough once unlocked.
Bat Tamer	400GP	Turns damage from all bats to a constant .5 HP damage	Rescue Riko. Can only be purchased once for each playthrough once unlocked.
Air Shoes	500GP	Allows Yohane to walk over water tiles. Prevents damage from standing on heat tiles.	Rescue You. Can only be purchased once for each playthrough once unlocked
Stewshine	1000GP	Increases Yohane's HP by 1 permanently	Rescue Mari. Can only be purchased once for each playthrough once unlocked
Mikan Mochi	1000GP	Increases Yohane's HP by 1 permanently	Rescue Chika. Can only be purchased once for each playthrough once unlocked
Kurosawa Macha	1000GP	Increases Yohane's HP by 1 permanently	Rescue Dia. Can only be purchased once for each playthrough once unlocked
Choco-Mint Ice Cream	2000GP	Saves Yohane from a fatal hit when on hand. Heals Yohane to full health afterwards	Rescue Ruby. Can only be purchased once for each playthrough once unlocked

Hanamaru: Yohane-chan, zura! what can I do for you today?

Total Gold: 365 GP

[1] Tears of a fallen Angel	30GP	[R]eturn
[2] Noppo Bread	100GP	

Choice:

Viewing Inventory

Players are also able to keep track of the items they have from the main menu. The only items that can be tracked are the Tears of a fallen angel, Noppo bread, and Choco-mint ice cream.

Lailaps: These are the items you have, Yohane!

HP: 4 / 5

Total Gold: 365 GP

Items available

1. Tears of a fallen angel	x	0
2. Noppo bread	x	2
3. Choco-mint ice cream	x	0

[R]eturn

Choice:

The Final Dungeon / Siren Battle

Once Yohane clears all 3 dungeons during the playthrough, the choices of which dungeon to travel to will be replaced with the singular option of facing the Siren

Lailaps: Yohane! It's time to face the Siren!

HP: 2 / 5

Item on hand: Tears of a Fallen Angel

Total Gold: 280 GP

[1] Face the Siren of Numazu

[I]nventory

[S]ave and Quit

[H]anamaru's Shop

choice:

The layout of the Siren's dungeon is the same, but the placement of the switches will be random. In the battle, the player will be controlling BOTH Yohane AND Lailaps

Final Battle: Siren of the Mirror world!

HP: 2 / 5

Lailaps HP: 4 / 4

Total Gold: 280 GP

Item on hand: Tears of a Fallen Angel

A 20x20 grid of dots. The top and bottom rows are filled with asterisks. The left and right sides are filled with a mix of asterisks and dots. The following characters are placed within the grid:

- Two '0' characters in the 5th row, 1st and 3rd columns.
- A red 'S' character in the 2nd row, 10th column.
- A cyan 'YL' character in the 18th row, 12th column.
- A single dot in the 12th row, 12th column.

During the final battle, Yohane and Lailaps spawn beside each other on a fixed location, always at the bottom of the dungeon. Here, the player is able to move both Lailaps and Yohane simultaneously. This means that, if Yohane moves up, Lailaps will also move up. If Yohane moves right, Lailaps also moves right. If, there is an obstacle blocking the character's movement, then only the character free from an obstacle will be able to move. An example will be if the player makes Yohane move up, but there is a wall blocking Lailap's way up, then only Yohane will move up and Lailaps will stay still.

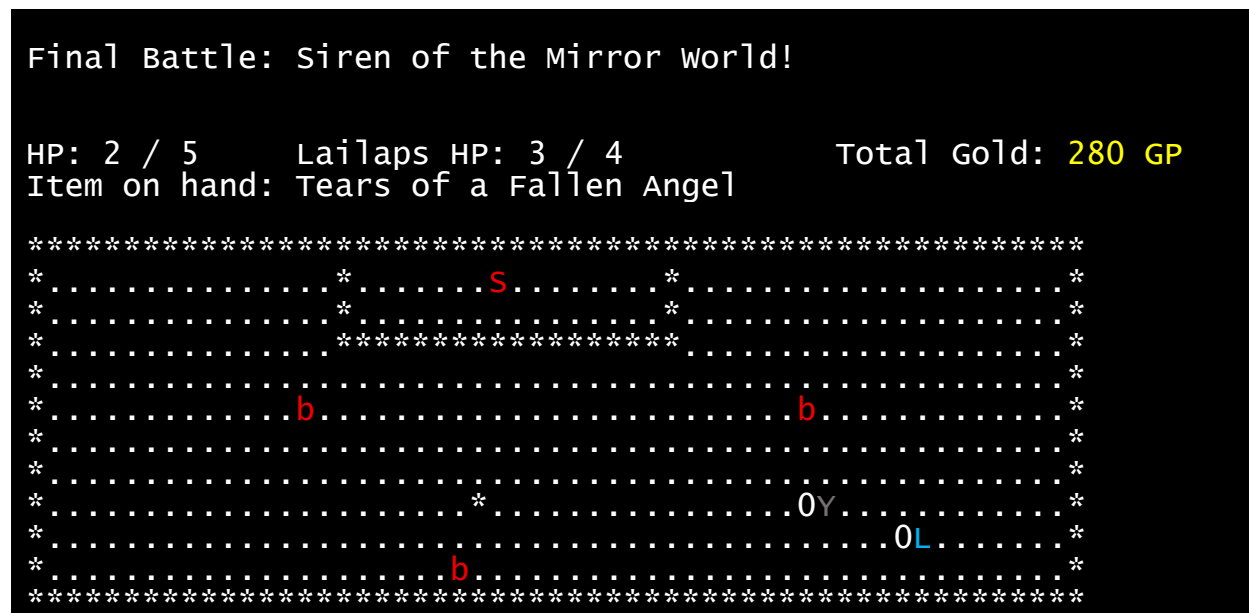
The flow of the battle is as follows:

- 1.) At the start of the battle, two switches (represented with the character 'O') will spawn in two different locations.
- 2.) The player must use the impassable tiles ('*') to position Yohane and Lailaps to match the position of the switches
- 3.) Yohane and Lailaps must then trigger both switches simultaneously
- 4.) The switches will then disappear and another new pair of switches will spawn elsewhere
- 5.) This process is repeated a total of 3 times, at which point, the walls surrounding the Siren will break free
- 6.) The Siren will then move towards Yohane and Lailaps, and can move diagonally.
- 7.) If Yohane can land a successful attack against the Siren, the Siren disappears (and leaves behind 750gp) and an exit is spawned in the Siren's original place. Moving to this exit allows the player to win the playthrough!
- 8.) Otherwise, the Siren will reduce Yohane OR Lailap's HP to 0 in one hit (Unless Yohane has Ruby's ice cream, in which case she'll survive one hit, but will have her HP reduced to 0 if she gets hit again). Likewise, Lailaps CANNOT attack, so if the Siren lands a hit on Lailaps, it's game over!

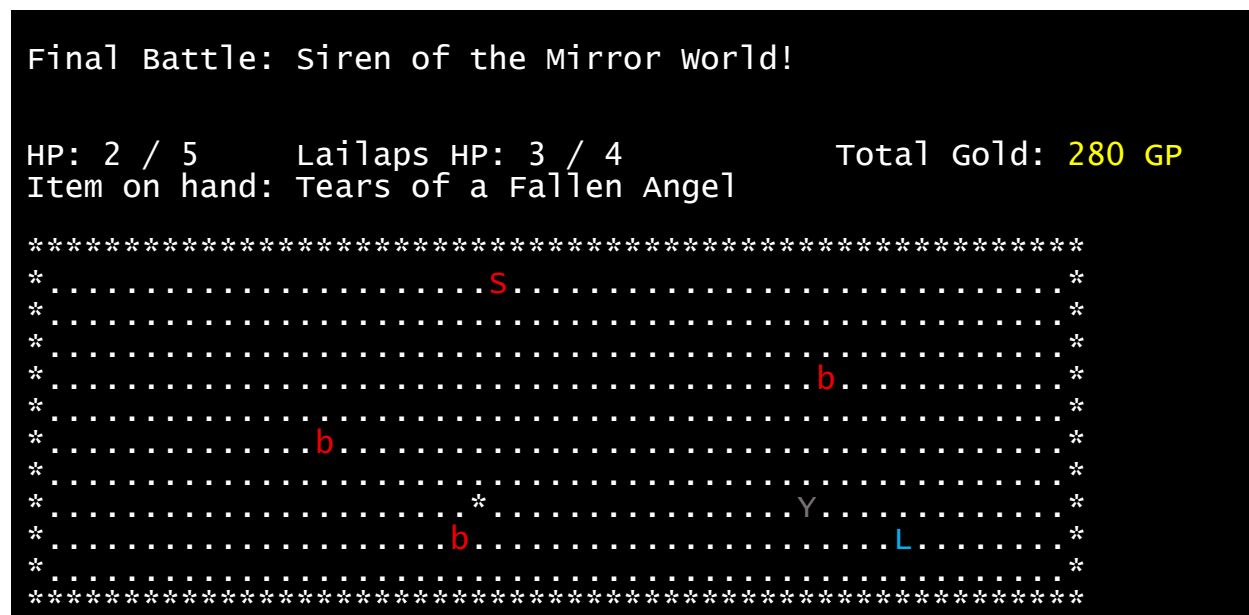
In the midst of Yohane and Lailaps trying to activate the switches, the Siren will summon a bat enemy every 8 moves. The Bat will spawn randomly on a passable tile that is not occupied by a switch. The bats will function depending on the number of successful switch triggers Yohane and Lailaps perform. 0 switch triggers make bats act like they are in the 1st dungeon, 1 switch trigger makes bats act like they are in the 2nd dungeon, and 2 switch triggers make bats act like they are in the 3rd dungeon

NOTE: When Spawning the switches, the pair of switches should NOT be further than two rows and five columns away.

The following set of images demonstrate the moment Yohane and Lailaps trigger the switches for the third time, thereby releasing the Siren's barriers.



Next move: Yohane (and Lailaps) moves left.



The barriers disappear once the switches are triggered for the third time. The Siren stops spawning in bats and in the next move, she will start moving towards Yohane.

Next move: Yohane & Lailaps move up, Siren moves diagonally SE

Final Battle: Siren of the Mirror world!

HP: 2 / 5 Lailaps HP: 3 / 4 Total Gold: 280 GP
Item on hand: Tears of a Fallen Angel

```
*****
*
* . . . . .
* . . . . . S . . . . .
* . . . . . . . . . . b . . . . .
* . . . . . . . . . . . . . . .
* . . . . . b . . . . . . . . . .
* . . . . . . . . . . . . . . .
* . . . . . . . . . . . . . . .
* . . . . . . . . . . * . . . . Y . . . .
* . . . . . . . . . . b . . . . . L . . . .
* . . . . . . . . . . . . . . .
*****
```

Next move: Yohane & Lailaps move up. Siren moves SE.

Final Battle: Siren of the Mirror world!

HP: 2 / 5 Lailaps HP: 3 / 4 Total Gold: 280 GP
Item on hand: Tears of a Fallen Angel

```
*****
*
* . . . . . . . . . . . . . . .
* . . . . . . . . . . . . . . b . . . .
* . . . . . . . . . . S . . . . .
* . . . . . b . . . . . . . . . .
* . . . . . . . . . . . . . . .
* . . . . . . . . . . . . . . .
* . . . . . . . . . . . . . . Y . . . .
* . . . . . . . . . . b * . . . . . L . . .
* . . . . . . . . . . . . . . .
* . . . . . . . . . . . . . . .
*****
```

Next move: Yohane & Lailaps move up again. Siren moves SE.

```
Final Battle: Siren of the Mirror world!

HP: 2 / 5      Lailaps HP: 3 / 4      Total Gold: 280 GP
Item on hand: Tears of a Fallen Angel

*****
* .....*
* .....*
* .....*
* .....b.....b.....*
* .....S.....*
* .....*
* .....Y.....*
* .....b.....L.....*
* .....*.....*
* .....*
* .....*
*****
```

Next move: Yohane and Lailaps move up. Siren continues to move SE.

```
Final Battle: Siren of the Mirror world!

HP: 2 / 5      Lailaps HP: 3 / 4      Total Gold: 280 GP
Item on hand: Tears of a Fallen Angel

*****
* .....*
* .....b.....b.....*
* .....*
* .....S.....Y.....*
* .....b.....L.....*
* .....*.....*
* .....*
* .....*
*****
```

Next move: Yohane and Lailaps move left. Now that the Siren is on the same row as Yohane, it now move right

Final Battle: Siren of the Mirror world!

HP: 2 / 5 Lailaps HP: 3 / 4 Total Gold: 280 GP
Item on hand: Tears of a Fallen Angel

```
*****
* .....b.....b.....*
* .....*.....*.....*
* .....*.....*.....*
* .....*.....*.....*
* .....b.....s.....Y.....*
* .....*.....*.....L.....*
* .....*.....*.....*.....*
* .....*.....*.....*.....*
* .....*.....*.....*.....*
* .....*.....*.....*.....*
* .....*.....*.....*.....*
*****
```

Next move: Yohane and Lailaps moves left. Siren moves right.

Final Battle: Siren of the Mirror world!

HP: 2 / 5 Lailaps HP: 3 / 4 Total Gold: 280 GP
Item on hand: Tears of a Fallen Angel

```
*****
* .....b.....b.....*
* .....b.....*.....*.....*
* .....*.....*.....*.....*
* .....*.....s.....Y.....*
* .....*.....*.....L.....*
* .....*.....*.....*.....*
* .....*.....*.....*.....*
* .....*.....*.....*.....*
* .....*.....*.....*.....*
* .....*.....*.....*.....*
* .....*.....*.....*.....*
*****
```

Next move: Yohane and Lailaps move left again. Siren continues to move right.

Final Battle: Siren of the Mirror world!

HP: 2 / 5 Lailaps HP: 3 / 4
Item on hand: Tears of a Fallen Angel

Total Gold: 280 GP

```
*****
* .....b.....b.....*
* .....*.....*.....*
* .....*.....*.....*
* .....S.Y.....*
* .....L.....*
* .....b.....*
* .....*.....*
* .....*.....*
* .....*.....*
* .....*.....*
* .....*.....*
*****
```

NOTE: At this current position, If Yohane moves left, that will leave her in a tile adjacent to the Siren. This will mean that the Siren WILL hit Yohane. As Yohane does NOT have the choco-mint ice cream on hand, that will lead to a game over.

Thus, the safe route would be either to move down, move right, or move up.

Next move: Yohane and Lailaps move up. Siren moves NE as a result.

```

Final Battle: Siren of the Mirror world!

HP: 2 / 5      Lailaps HP: 3 / 4      Total Gold: 280 GP
Item on hand: Tears of a Fallen Angel

*****
*
*      b      b
*      sY
*      L
*
*
*      b
*      *
*
*
*****

```

Now Yohane and the Siren are adjacent to each other. The correct move here is to press left and attack the Siren. If Yohane moves Up or Down, the Siren will be able to hit her, reducing her HP to 0 and causing a game over. Moving right, however, is safe, but the Siren will just move right again, and doing so can cause the bat above Lailaps to riskily move adjacent to her.

Next move: Yohane attacks the siren (Presses left), Lailaps moves left as a result, Siren defeated!

Final Battle: Siren of the Mirror world!

HP: 2 / 5 Lailaps HP: 3 / 4
Item on hand: Tears of a Fallen Angel

Total Gold: 280 GP

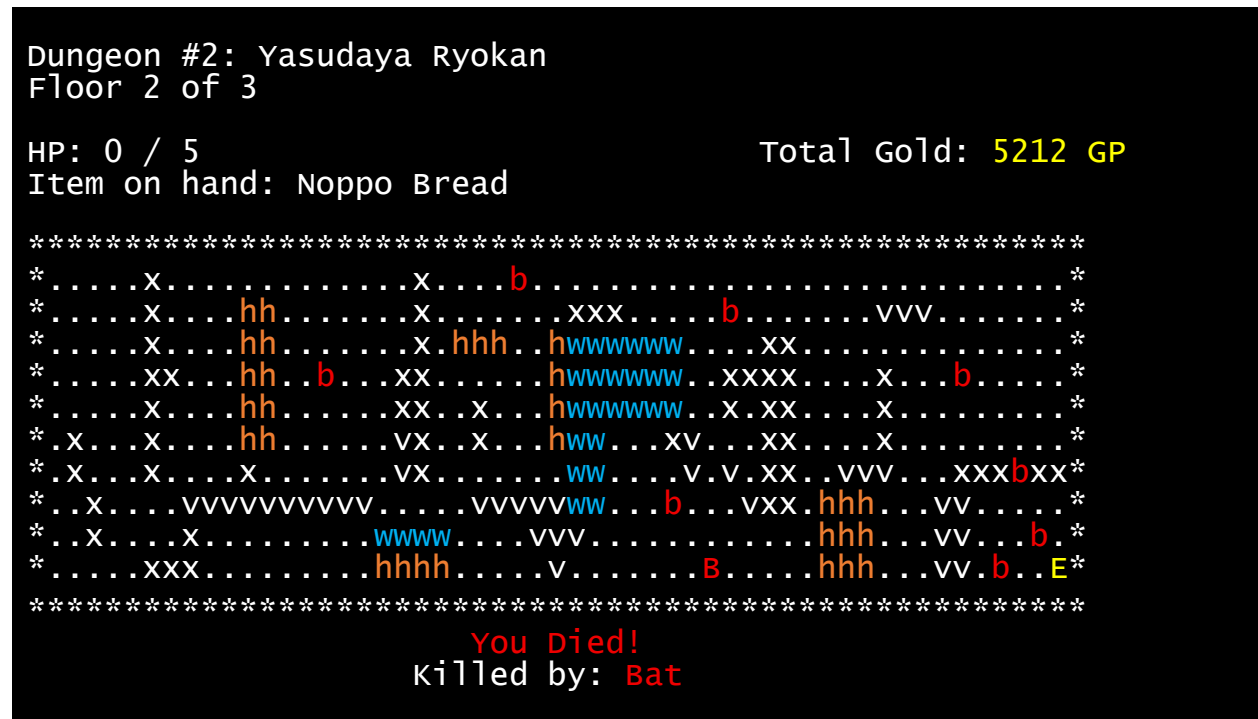
```
*****
*               E               *
*   g               g               *
*               gY               *
*               L               *
*               *               *
*   g               *               *
*               *               *
*               *               *
*               *               *
*               *               *
*               *               *
*               *               *
*               *               *
*               *               *
*****
```

Pressing left causes Yohane to attack the Siren, defeating it. Lailaps, however, moves left as there is nothing obstructing her movement.

With the defeat of the siren, all the bats fall and the Siren leaves behind 750gp from her location. The Exit spawns and the player ends up winning the game!

Game Over

A player encounters a game over if Yohane's HP (or Lailaps' during the final boss battle) hits 0. The player is informed that they got a game over and what ended up killing them.



The following are obstacles that can considerably cause a game over:

- 1.) Bat
- 2.) Heat Tiles
- 3.) Spike Walls
- 4.) Siren

Should a player end up with a Game Over, the following are the things that are saved and not lost:

- 1.) The Aqours members that have been rescued in the playthrough (and previous playthroughs)
- 2.) Gold obtained
- 3.) Progress towards achievements
- 4.) Amount of Noppo-bread and Tears of a Fallen Angel

The following are the things that reset during the next playthrough

- 1.) The number of Aqours members
- 2.) The upgrades purchased in that playthrough
- 3.) Maximum HP is reset back to 3

Status

The status page keeps track of the player's progress throughout the entirety of the game. In particular, it lists the number of times each idol has been saved, the number of gold spent, the number of runs won, and the number of times they have encountered a game over. See the sample screen shown below in the console edition:

```
*****
                        overall Status
*****
Times Chika was saved           0 times
Times You was saved             1 times
Times Riko was saved            0 times
Times Hanamaru was saved        1 times
Times Ruby was saved            1 times
Times Dia was saved             1 times
Times Kanan was saved           0 times
Times Mari was saved            1 times

Times Siren was defeated        2 times

No. of game overs               4 times

Total gold spent                4872 gp
```

Full Song Edition +++

For this, you are going to implement it in two phases. The first phase is implemented in Console using Java. The second phase requires implementing the FULL Game with GUI. More of this in the next section

First Phase: Only 1 dungeon required

Second Phase: 3 dungeons AND the Final Boss Battle With GUI

Character Profile(s)

Name	Description	Appearance	Appearance (Mirror World)
Yoshiko Tsushima (Yohane)	Yoshiko is a 1 st year member of the idol group Aqours, who claims she's a fallen angel who was cast out of heaven. This persona enabled her alternate self in the mirror world to possess magical powers, capable of protecting Numazu from supernatural threats. <i>cv. Aika Kobayashi</i>		
Lailaps	Lailaps exists in the mirror world as Yohane's loyal wolf companion. She acts as the voice of reason and guides Yohane throughout her mission. <i>cv. Yoko Hikasa</i>		
Chika Takami	Chika is a 2 nd year and the founding member of Aqours. Her family runs a Ryokan inn, similarly to her mirror world self, where she specializes in her family's signature Mikan Mochi! <i>cv. Anju Inami</i>		
You Watanabe	You is a 2 nd year member of Aqours, who is Chika's childhood friend well-known for her athletic prowess. In the mirror world, she's a post-lady with equipment that allows her to travel far and wide through the land of Numazu. <i>cv. Saito Shuka</i>		

Riko Sakurauchi	Riko is a 2nd year member of Aqours who transferred to Uchiura from Tokyo. As the pianist and main composer of the group, her diligence carries over to the mirror world where she acts as a zoologist capable of understanding and manipulating beast behavior. <i>cv. Rikako Aida</i>		
Hanamaru Kunikida	Hanamaru is a 1st year member of Aqours, well-known for her fondness for food, books, and her tendency to say “zura” at the end of every sentence. She’s a travelling merchant in the mirror world who specializes in selling sweets and all sorts of goodies to help adventurers along their journey. <i>cv. Kanako Takatsuki</i>		
Ruby Kurosawa	Ruby is a 1st year member of Aqours, who is Hanamaru’s childhood friend. This carries over to the mirror world where Ruby, now a fairy, helps Hanamaru by supplying equipment to her shop. She also bestows magic in her sweet treats, some of which is rumored to be “life-savingly” delicious. <i>cv. Ai Furihata</i>		
Dia Kurosawa	Dia is a 3rd year member of Aqours and is Ruby’s older sister. As she’s the student council president of Uranohoshi high school, her mirror world self is the chief of staff of the Numazu Administrative Bureau. However, her mirror self is known to be		

	highly fond of anything macha. <i>cv. Arisa Komiya</i>		
Kanan Matsuura	Kanan is a 3 rd year member of Aqours, who runs a dive shop with her family. Her mirror world self is a mechanic who creates powerful and useful gadgets to make working and performing tasks in Numazu more convenient for citizens. <i>cv. Nanaka Suwa</i>		
Mari Ohara	Mari is a 3 rd year member of Aqours. As part of an affluent and wealthy family, she has been designated as the school director of Uranohoshi high school. Her mirror self is no different, being a demon lord who commands a small horde of friendly creatures in the island of Awashima <i>cv. Aina Suzuki</i>		
Siren of the Mirror World	Nobody has seen her and lived to tell the tale. It has been rumored that the Siren sings with a voice that is irresistible enough to lure unsuspecting individuals to their doom. Recently, this Siren has attempted to go beyond her reality and influence even the real world, starting with taking voices of idols all around the world in an attempt to enhance her own voice	????????????????????	????????????????????

CCPROG3 MP Directives

I. Milestones (see section II for Deliverables)

MCO1 – due July 11, 9PM

1. Draw the UML class diagram to represent the system described.
2. Implement only the classes REQUIRED for the completion of just 1 Dungeon. For this phase, the idol to be rescued CAN BE FIXED (1 of the 8 possible idols) and NOT RANDOM. Hanamaru's store is NOT yet required to be implemented.
3. Phase I requires the output to be displayed on the console
4. Save and Quit NOT YET NEEDED to be implemented

MCO2 – due August 3, 7:30AM

1. You must follow the MVC architecture.
2. Your final system must be a GUI-based program. All interactions will be on the GUI components. Console outputs will not be checked. This means:
 1. Rendering of the 2D Map must be done through images and sprites
 2. Controlling Yohane may be done via Keyboard Events OR interactable buttons on the GUI (Up, Down, Left, Right Button)
 3. The main menu can be in the form of an interactable GUI (button presses) OR an interactive 2D Map
3. Complete UML class diagram.
4. Phase 2 implementation requires GUI.
5. Additional features may be implemented by the programmer. However, these should not contradict the requirement.
6. Implemented additional features should be reflected in the UML class diagram too.

II. Deliverables

The deliverables for both MCOs include:

1. The design and implementation of the solution should...
 - Conforms to the specifications described above
 - Exhibit proper object-based / object-oriented concepts, like encapsulation and information-hiding, etc.
 - NOT be derived or influenced from any use of Generative AI tools or applications
2. To allow for an easier time to validate the program, usage of libraries outside of what is available in the Java 21 API is not allowed.
3. Signed declaration of original work (declaration of sources and citations may also be placed here)
 - See Appendix A for an example
4. Softcopy of the class diagram following UML notations (in pdf or png)
 - Ensure that the diagram is easy to read and well structured

5. Javadoc-generated documentation for proponent-defined classes with pertinent information
6. Zip file containing the source code with proper internal documentation
 - The program must be written in Java
7. Test script following the format indicated in Appendix B
 - In general, there should be at least 3 categories (as indicated in the description) of test cases per method (except for setters and getters).
 - There is no need to test user-defined methods which are ONLY for screen design (i.e., no computations/processing; just print/println).
8. For MCO1 only: A video demonstration of your program (max 10 minutes)
 - While groups have the freedom to conduct their demonstration, a demo script will be provided closer to the due date to help with showing the expected functionalities.
 - The demonstration should also quickly explain key aspects of the program's design found in the group's class diagram
 - Please keep the demo as concise as possible and refrain from adding unnecessary information
 - Video demonstration should include voice-over explanation (not music background nor pure silence)
9. The student/s should make pertinent back-ups of his/her/their own project. A softcopy of the final unmodified files (for each phase) should be sent to the student/s own email address/es, apart from regular submissions of progress in AnimoSpace and/or Git.

III. Submission

All deliverables for the MCO are to be submitted via AnimoSpace. Submissions made in other venues will not be accepted. Please also make sure to take note of the deadlines specified on AnimoSpace. No late submissions will be accepted.

II. Grading

For grading of the MCO, please refer to the MCO rubrics indicated in the syllabus.

III. Collaboration and Academic Honesty

This project is meant to be worked on as a pair (i.e. max of 2 members in a group). In exceptional cases, a student may be allowed by their instructor to work on the project alone; however, permission should be sought as collaboration is a key component of the learning experience. Under no circumstance will a group be allowed to work on the MCO with more than 2 members.

A student cannot discuss or ask about design or implementation with other persons, with the exception of the teacher and their groupmate. [Questions about the MP specs should be raised in the Discussion page in AnimoSpace] Copying other people's work and/or working in collaboration with other teams are not allowed and are punishable

by a grade of 0.0 for the entire CCPROG3 course and a case may be filed with the Discipline Office. In short, do not risk it; the consequences are not worth the reward. Comply with the policies on collaboration and AI usage as discussed in the course syllabus.

IV. Documentation and Coding Standards

Do not forget to include internal documentation (comments) in your code. At the very least, there should be an introductory comment and a comment before every class, every method, and every attribute. This will be used later to generate the required External Documentation for your Machine Project via javadoc. You may use an IDE or the appropriate command-based instructions to create the documentation, but it must be PROPERLY constructed.

Please note that we're not expecting you to add comments for each and every line of code. A well-documented program also implies that coding standards are adhered to in such a way that they aid in the documentation of the code. Comments should be considered for more complex logic.

V. Bonus Points

No bonus points will be awarded for MC01. Bonus points will only be awarded for MC02. The above description of the program is the basic requirement. Any additional feature will be left to the creativity of the student. Bonus points would be awarded depending on the additional implemented features. These additional features could include the following:

- Background Music
- New Enemy with new behavior (Skeletons, Dragons, Ghouls, etc.)
- Achievement Module
- Real Time Movement instead of Turn Based Movement (Bats move every second instead of waiting for Yohane's turn)

VI. Resources and Citations

All sources should have proper citations. Citations should be written using the APA format. Examples of APA-formatted citations can be seen in the References section of the syllabus. You're encouraged to use the declaration of original work document as the document to place the citations.

Further, this is to emphasize that you DO NOT need to create your own sprites (background pictures, animal pictures, etc.) for the game. You can just use what's available on the Internet and just include them into your project, just make sure to cite your sources.

VII. Demo

Demo for MCO1 is via a video submission. All members are expected to be present in the video demonstration and should have relatively equal parts in terms of the discussion. Any student who is not present during the demo will receive a zero for the phase.

In MCO2, demo is live and will include an individual demo problem. Schedule for the demo will generally be during class time, but there may be other schedules opened by the faculty in case there is not enough time to accommodate everyone. Sequence (of who goes first in the class to do the demo) is determined by the faculty. Thus, do not be absent or late during announced demo days. A student or a group who is not present during the demo or who cannot answer questions regarding the design and implementation of the submitted project convincingly will incur a grade of 0 for that project phase.

During the MP demo, it is expected that the program can be compiled successfully in the command prompt and will run. If the program does not run, the grade for that phase is 0. However, a running program with complete features may not necessarily get full credit, as design and implementation (i.e., code) will still be checked.

VIII. Other Notes

You are also required to create and use methods and classes whenever possible. Make sure to use Object-Based (for MCO1) and Object-Oriented (for MCO2) Programming concepts properly. No brute force solution.

Statements and methods not taught in class can be used in the implementation. However, these are left for the student to learn on his or her own.

Appendix A. Template for Declaration of Original Work

Declaration of Original Work

We/I, [Your Name(s)] of section [section], declare that the code, resources, and documents that we submitted for the [1st/2nd] phase of the major course output (MCO) for CCPROG3 are our own work and effort. We take full responsibility for the submission and understand the repercussions of committing academic dishonesty, as stated in the DLSU Student Handbook. We affirm that we have not used any unauthorized assistance or unfair means in completing this project.

[In case your project uses resources, like images, that were not created by your group.]
We acknowledge the following external sources or references used in the development of this project:

1. Author. Year. Title. Publisher. Link.
2. Author. Year. Title. Publisher. Link.
3. Author. Year. Title. Publisher. Link.

By signing this declaration, we affirm the authenticity and originality of our work.

<i>Signature and date</i>	<i>Signature and date</i>
Student 1 Name ID number	Student 2 Name ID number

[Note to students: Do not submit documents where your signatures are easily accessible. Ideally, submit a flattened PDF to add a layer of security for your digital signatures]

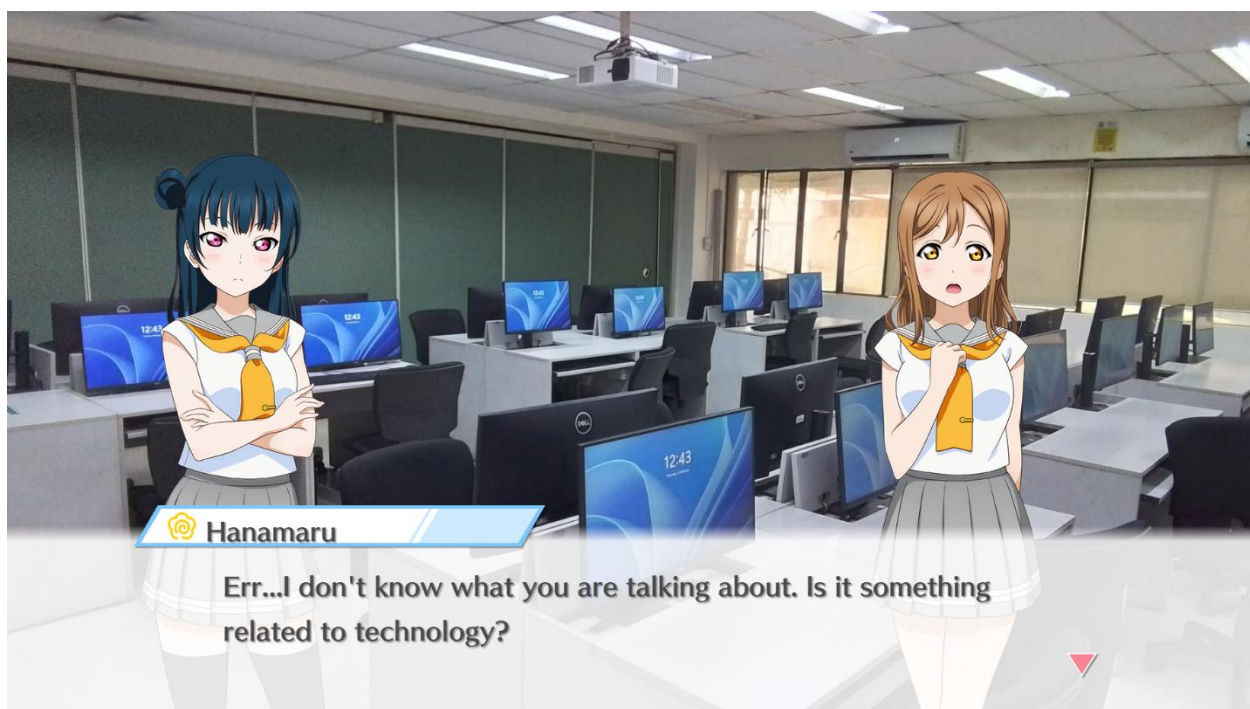
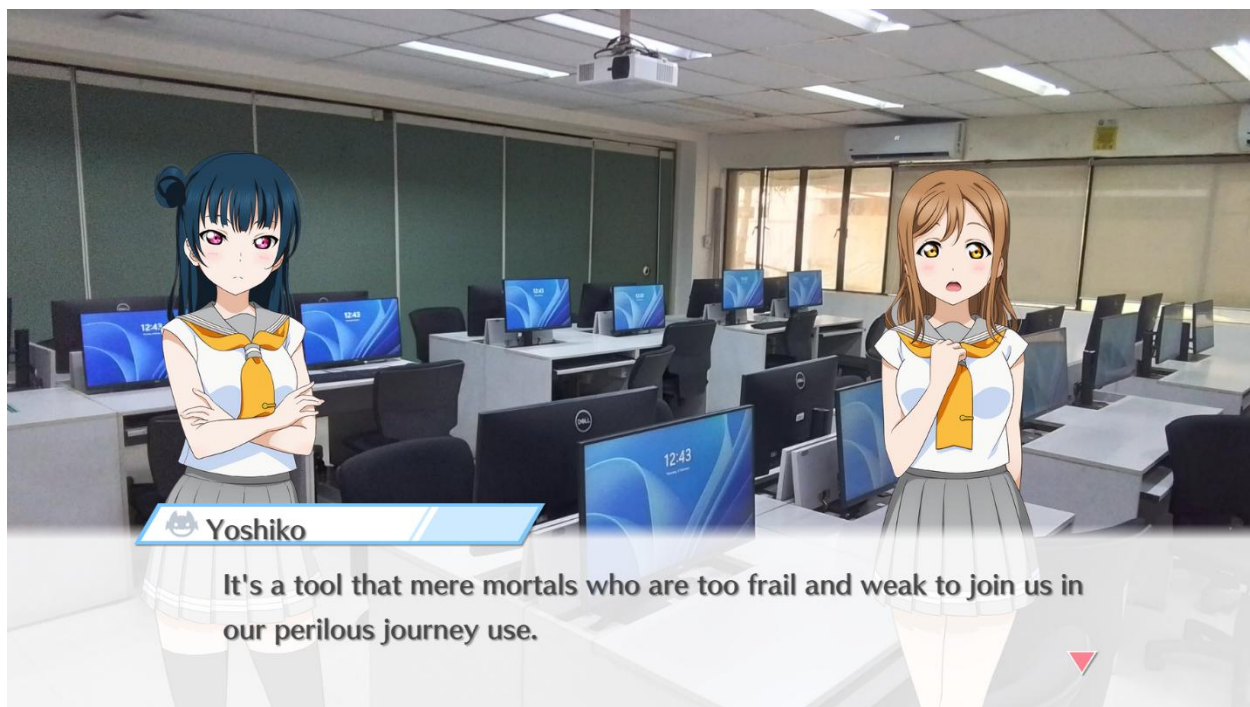
Appendix B. Example of Test Script Format

Class: MyClass						
Method	#	Test Description	Sample Input Data	Expected Output	Actual Output	P/F
isPositive	1	Determines that a positive whole number is positive	74	true	true	P
	2	Determines that a positive floating point number is positive	6.112	true	true	P
	3	Determines that a negative whole number is not positive	-871	false	false	P
	4	Determines that a negative floating point number is not positive	-0.0067	false	false	P
	5	Determines that 0 is not positive	0	false	false	P

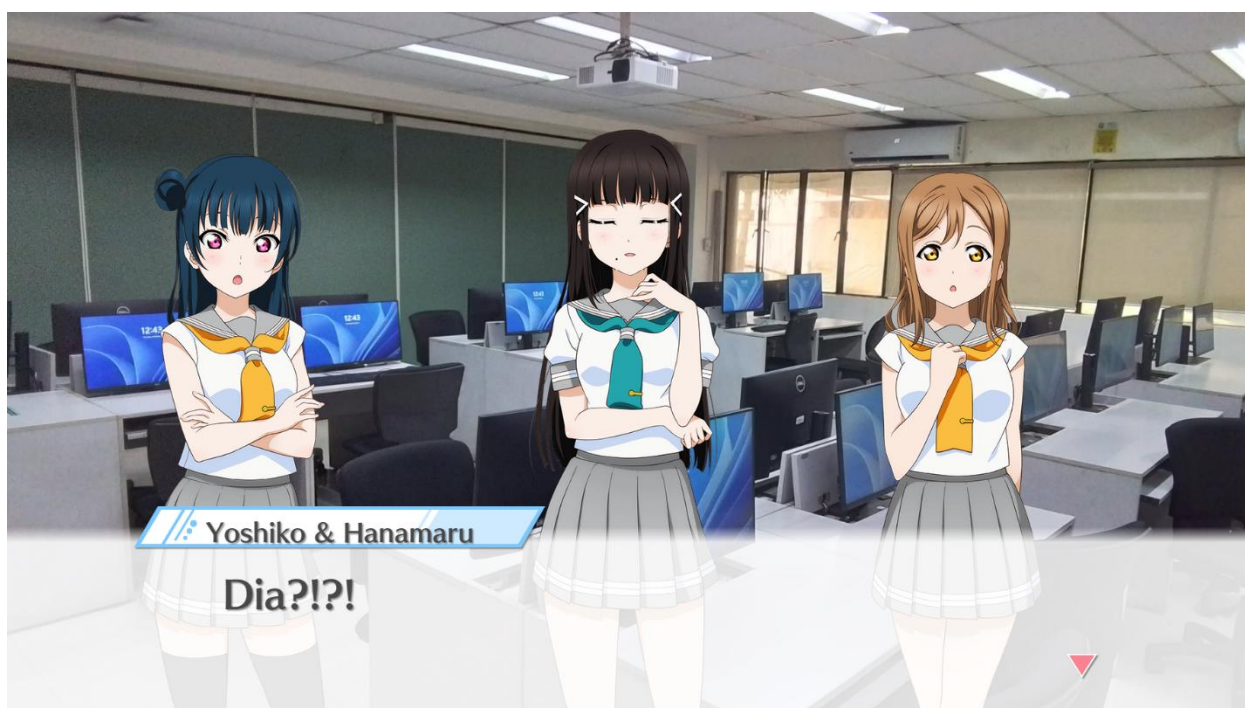
Likewise, using Generative AI for any modules within the codes in the project is counted as cheating. Codes used by Generative AI are a result of training data from other programmers, thus it's the same as using someone else's code that is not yours.

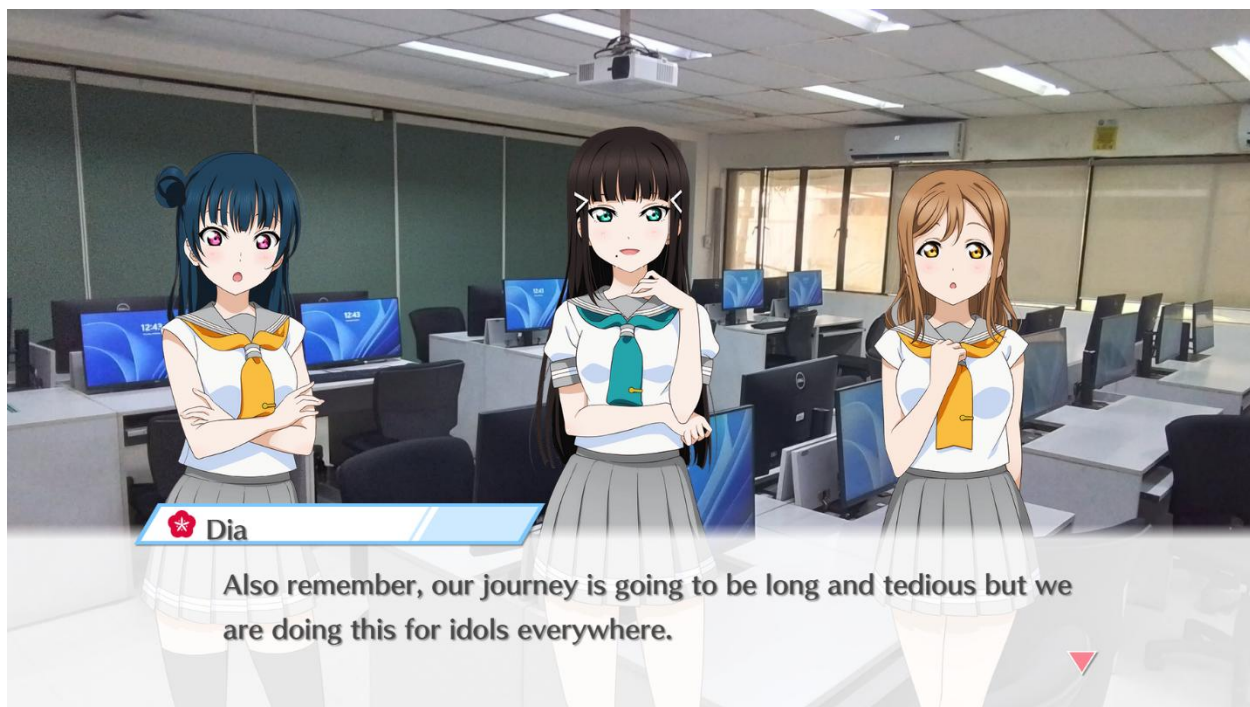
Bonus

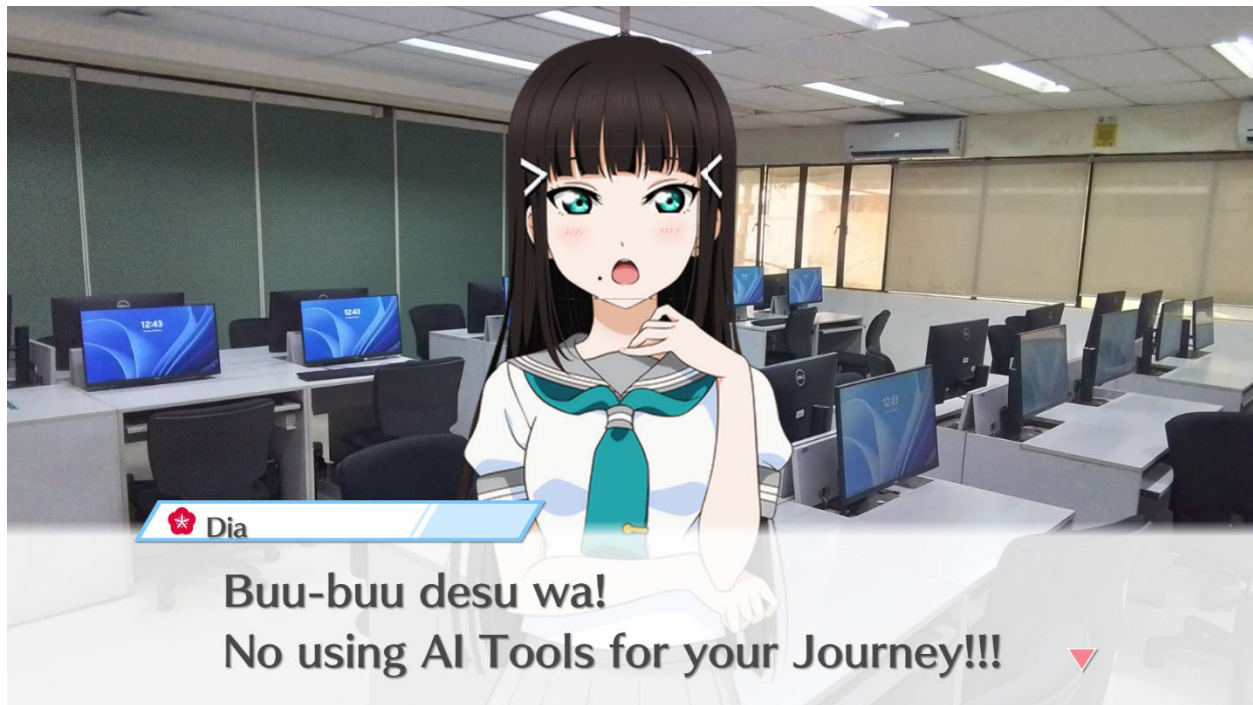












This project is dedicated to Aqours' 10th anniversary.

From 0 to 1 to 100, to 10 years, to eternal.